

Why eve and mallory love android: an analysis of android SSL (in)security

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Why eve and mallory love android: an analysis of android SSL (in)security

[Sascha Fahl](#), Marian Harbach, Thomas Muders, Matthew Smith, Lars Baumgärtner and Bernd Freisleben. the ACM Conference on Computer and Communications Security, CCS'12, Raleigh, NC, USA, October 16-18, 2012

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Abstract

Many Android apps have a legitimate need to communicate over the Internet and are then responsible for protecting potentially sensitive data during transit. This paper seeks to better understand the potential security threats posed by benign Android apps that use the SSL/TLS protocols to protect data they transmit. Since the lack of visual security indicators for SSL/TLS usage and the inadequate use of SSL/TLS can be exploited to launch Man-in-the-Middle (MITM) attacks, an analysis of 13,500 popular free apps downloaded from Google's Play Market is presented. We introduce MalloDroid, a tool to detect potential vulnerability against MITM attacks. Our analysis revealed that 1,074 (8.0%) of the apps examined contain SSL/TLS code that is potentially vulnerable to MITM attacks. Various forms of SSL/TLS misuse were discovered during a further manual audit of 100 selected apps that allowed us to successfully launch MITM attacks against 41 apps and gather a large variety of sensitive data. Furthermore, an online survey was conducted to evaluate users' perceptions of certificate warnings and HTTPS visual security

indicators in Android's browser, showing that half of the 754 participating users were not able to correctly judge whether their browser session was protected by SSL/TLS or not. We conclude by considering the implications of these findings and discuss several countermeasures with which these problems could be alleviated.

Reference

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